

LabView Reference Design Guide

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1 Figures and Tables

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Revision History

Version	Date	Notes
1.1	11.2015	Initial Release
1.2	02.2019	Minor corrections
1.3	05.2023	Update the operation instructions

2.1 Disclaimer

This reference design demonstrates usage of KAYA Instruments frame grabbers under LabView. KAYA Instruments assumes no responsibility for any damages resulting from the use of this product for purposes other than those stated.

Even if the product is used properly, KAYA Instruments assumes no responsibility for any damages caused by the following:

- Secondary impact arising from use of this product or its unusable state (business interruption or others).
- Use of this product against the instructions given in this manual or malfunctions due to connection to other devices.

KAYA Instruments assumes no responsibility or liability for:

- Erasure or corruption of data arising from use of this product.
- Any consequences or other abnormalities arising from use of this product, or damage of this product not due to our responsibility or failure due to modification.

3 Operation instructions

1. Open and run the main.lv file (using x64 bit LabView).
2. All the vi's will be updated to latest version of your LabView. Save them and restart the LabView with the updated design.
3. Make sure the Frame Grabber/Chameleon and camera are connected in the computer.
4. Select the required device from the "Frame Grabbers" dropdown menu
5. Select the required camera from the "Cameras" dropdown menu
6. Adjust the required "Image Width" and "Image Height".
7. Clicking the "Start" button will start the acquisition, clicking it again will stop it
8. In order to exit the application, click the "Exit" button.

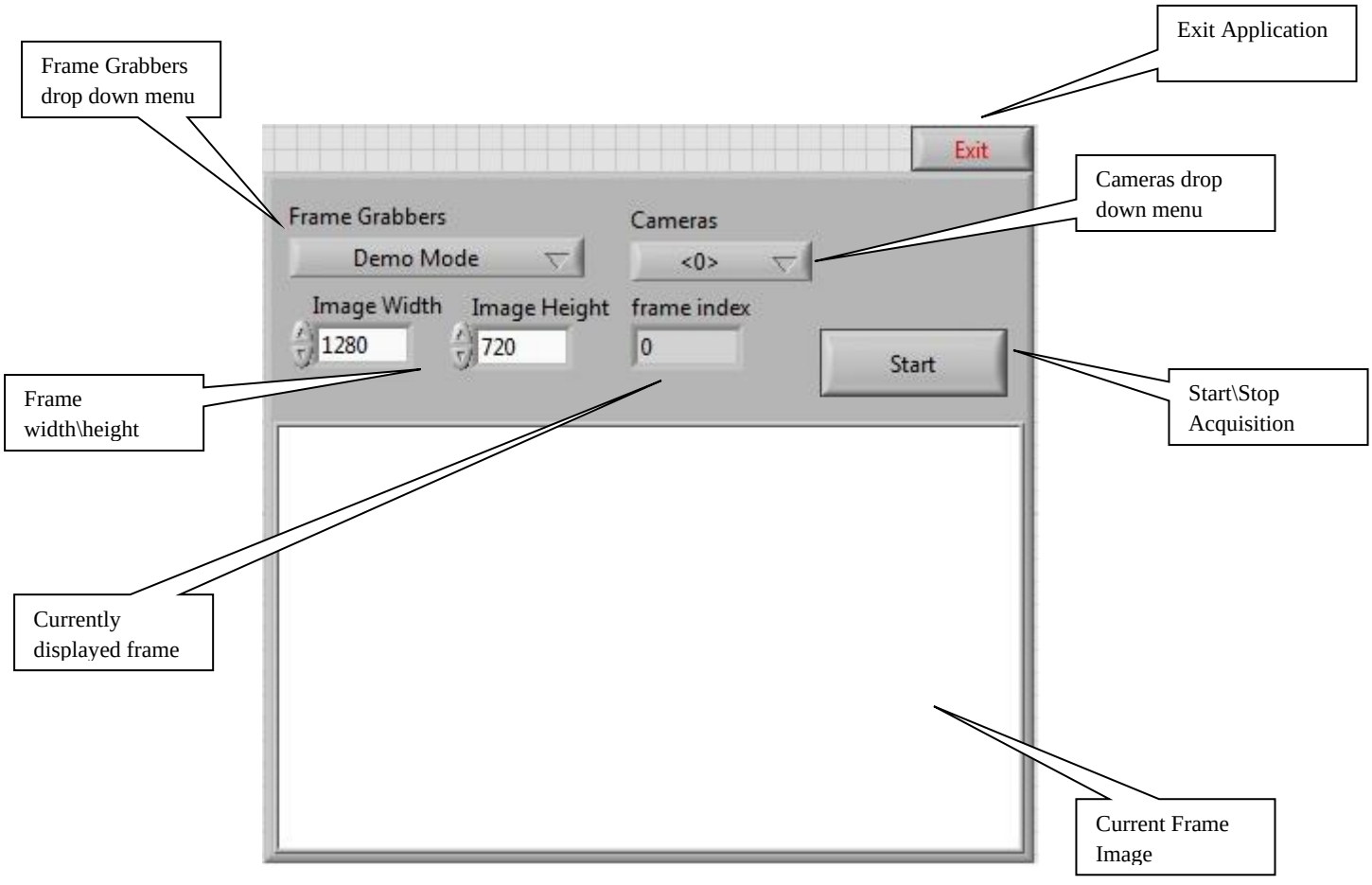


Figure 1: Main LabView window

4 Directory Structure

- Subvi : utility Vis' used for the application
- VI's : LabView wrapper Vis' for functions included in KYFGLib.dll, such as: KYFG Camera Open, KYFG Camera Start Acquire and KYFG Camera Stop Acquire
- KYFGLib_lv.lvlib : LabView library file for the project, includes all relevant files from the project directory
- main.vi: The main application file for frame acquisition.

1. When launching the application, it will automatically query the number of available Frame Grabbers, get their names and display them in the “Frame Grabbers” drop down menu after which it will automatically open a connection to the first available Frame Grabber.

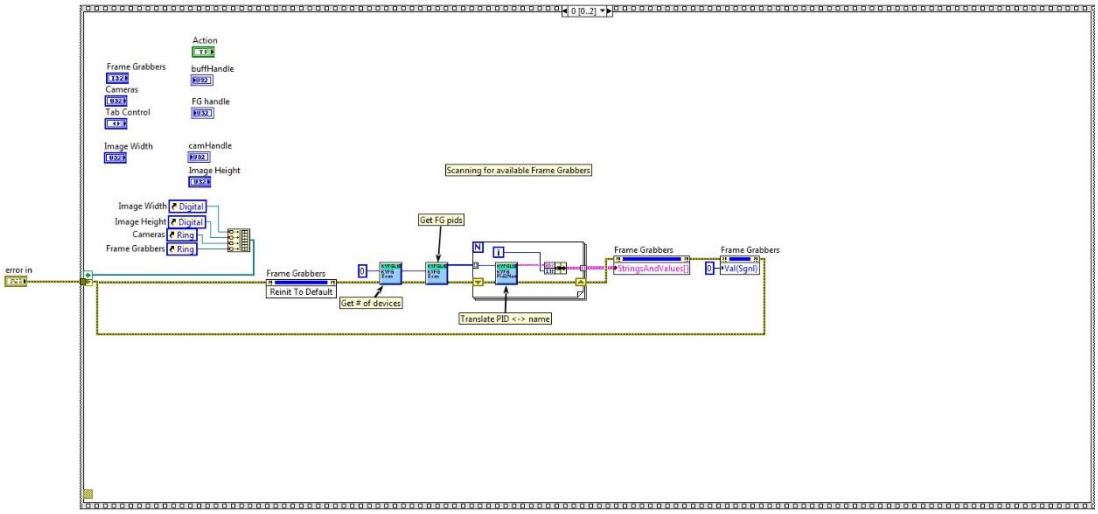


Figure 2: Launching the application

2. Selecting a new Frame Grabber from the “Frame Grabbers” drop down menu will close the previously opened Frame Grabber connection, open a new connection to the selected one, scan the Frame Grabber for cameras and update the “Cameras” drop down menu with the currently available number of cameras.

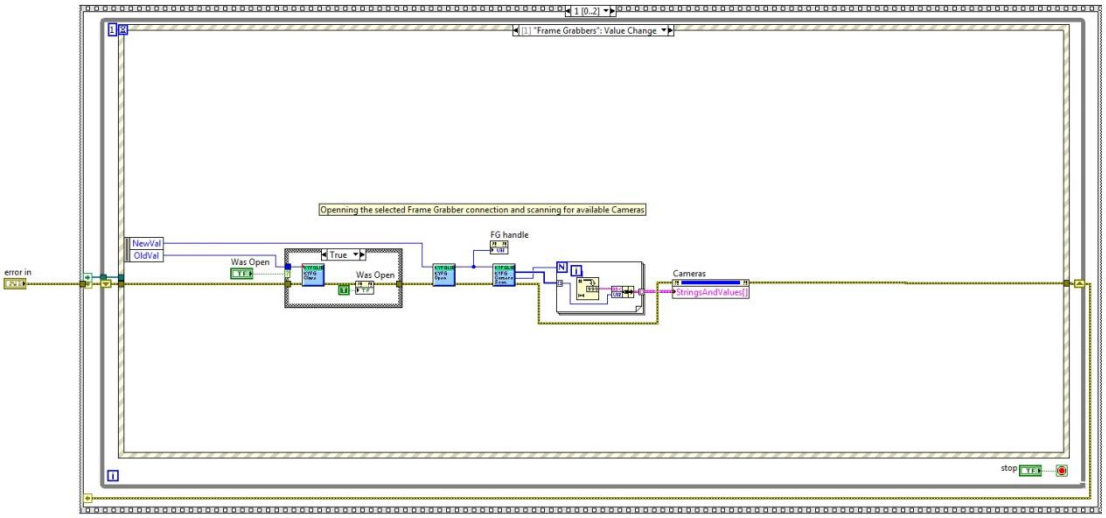


Figure 3: Selecting a new Frame Grabber

3. Selecting a camera from the “Cameras” drop down menu only updates a variable with the currently selected camera index.
4. Clicking the start button will disable all controls, and configure the following camera parameters:
    a. Pixel Format = Mono8
    b. Image Width = [Selected by user]
    c. Image Hight = [Selected by user]

After which a buffer will be allocated and the acquisition will start.

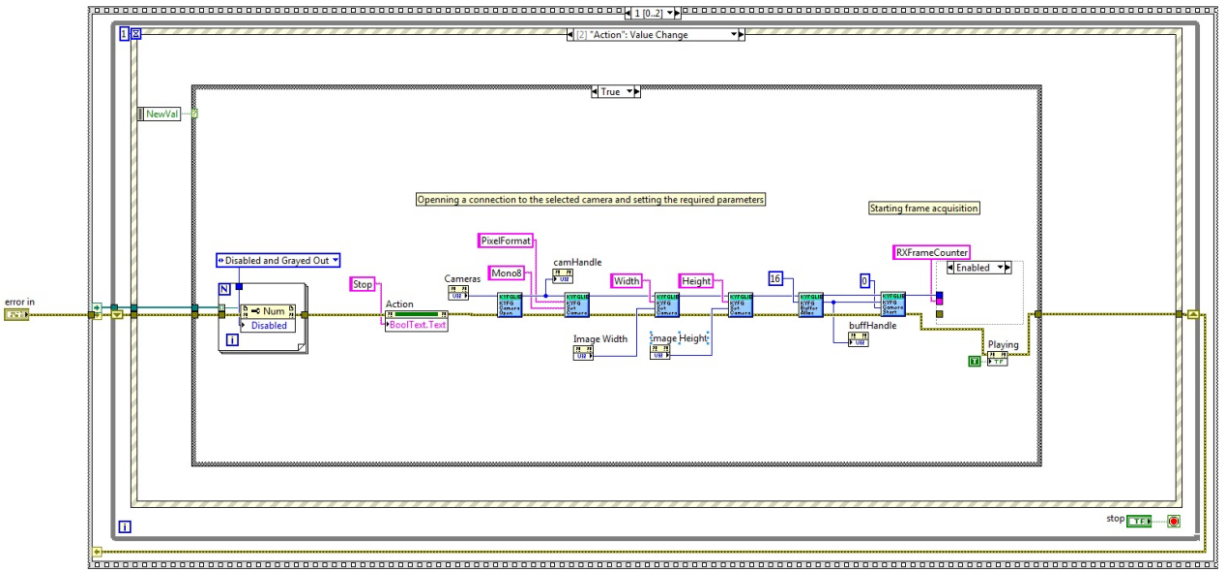


Figure 4: Activating the acquisition

In case the original image dimension of the camera should be used, need to disable the update block:

a. Open the Block diagram: “Window -> “Show Block Diagram”
b. In the Block diagram - Choose outer window “1” -> Choose scene “[3] Action : Value Change”
c. Change the camera resolution settings to read the resolution **from** the camera instead of setting it **to** the camera. This can be done by replacing the active vi’s with the disabled ones using “Enable this Sub diagram”

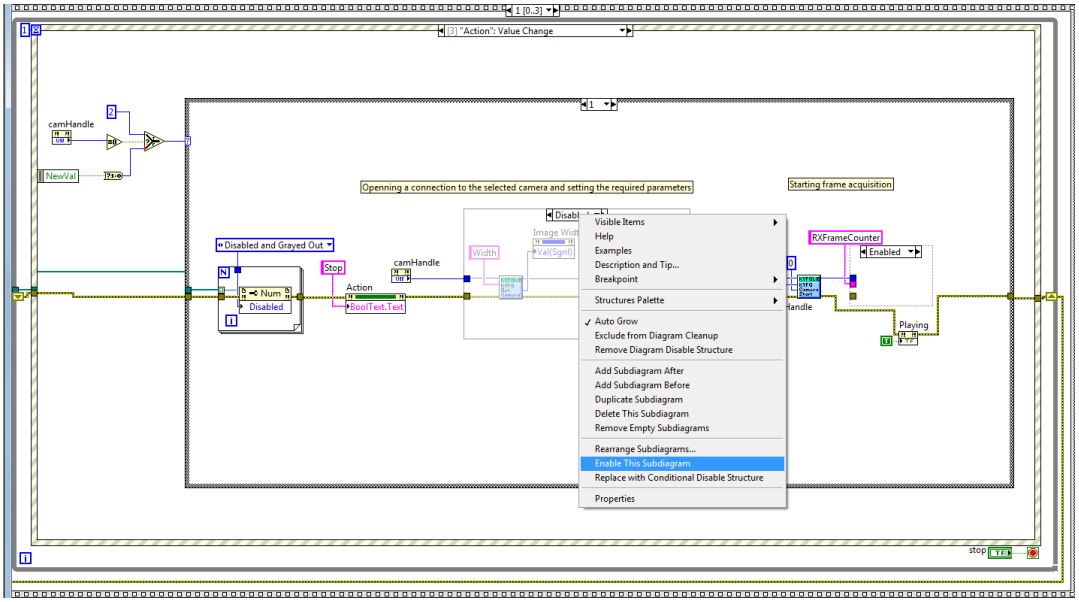


Figure 5: Enable dimensions change

5. When the acquisition is active the application will retrieve the current frame in the LabView “timeout event”. The application will first query the available buffer size and the current frame index, after which it will retrieve the pointer to the beginning of the buffer using “memcpy” to copy the buffer into a newly allocated array. Reshaping and resizing the retrieved 1D array into an image is required and done using native LabView function such as “reshape array” and draw image.

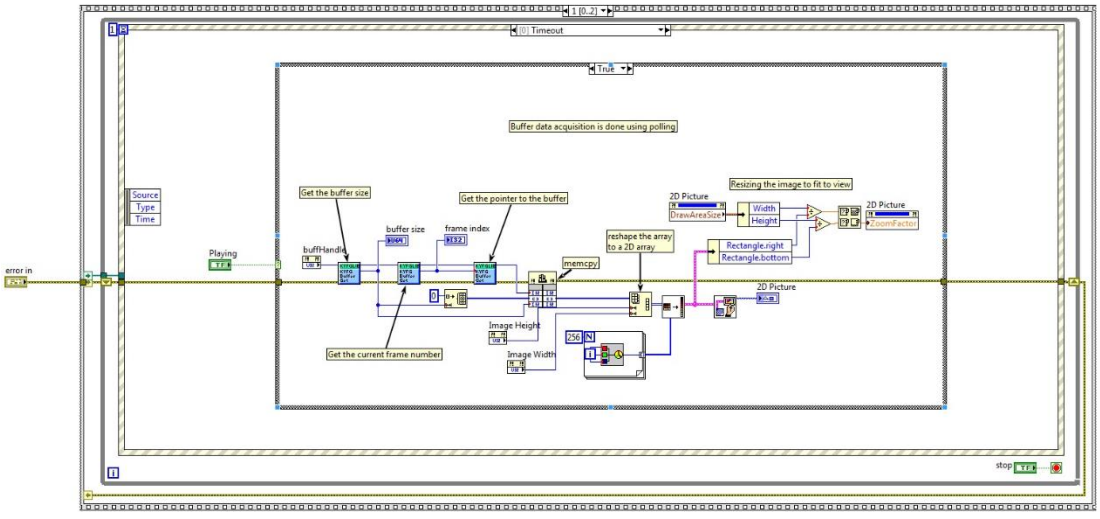


Figure 6: Reshaping and resizing

6. Clicking the “Stop” button will enable all previously disabled controls, stop the frame acquisition and close the handle to the selected camera.

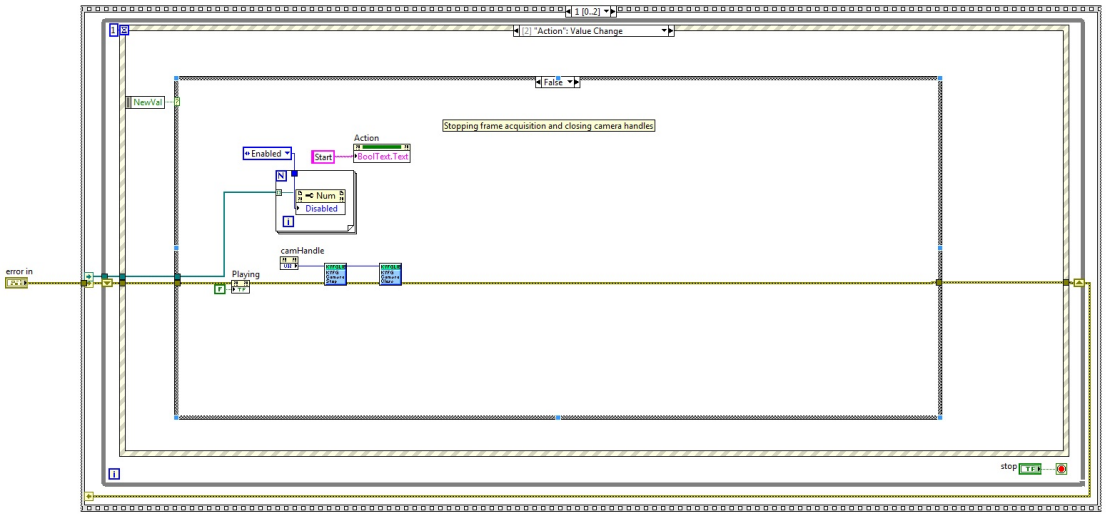


Figure 7: Closing the camera

7. Clicking the “Exit” button will close the handle to the currently selected Device and exit the application.

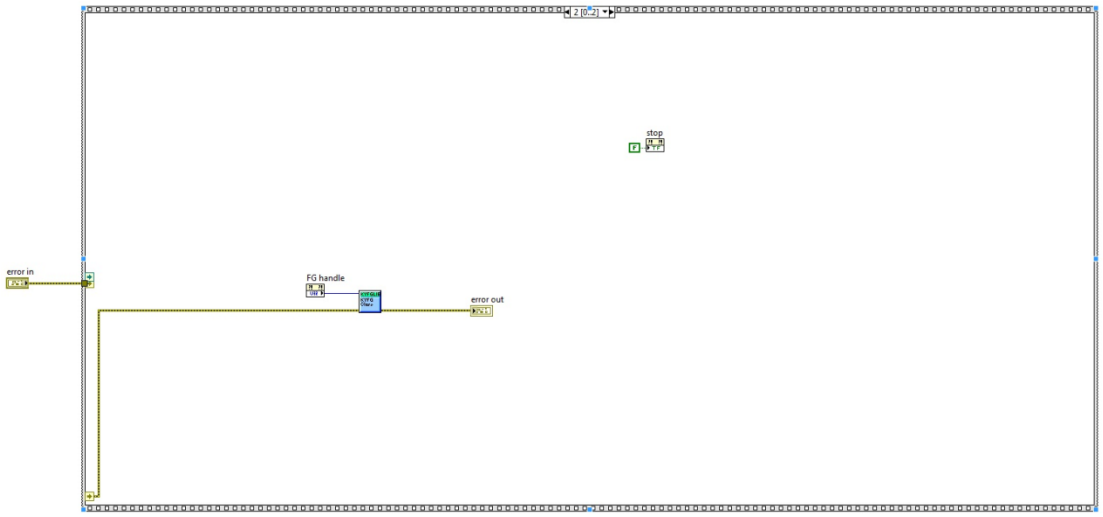


Figure 8: Exiting the application